



Chapter artwork

AGRICULTURE

Chapter 4 • Class 10 Geography

Complete handwritten-style revision notes covering the full chapter: farming systems, cropping seasons, major food and non-food crops, reforms, Bhoodan-Gramdan, exercises and activities.

CORE IDEA

Agriculture is a primary activity, a source of food, industrial raw material and export earnings. It depends on physical environment, technology and socio-cultural practices.



1. INDIA + AGRICULTURE: THE BIG PICTURE TYPES OF FARMING

- India is agriculturally important; about two-thirds of the population is engaged in agricultural activities.
- Agriculture is a primary activity: it provides most of the food we eat and raw material for many industries.
- Tea, coffee, spices and some other agricultural products are exported.

Industries can be based on agricultural raw material – e.g. food processing, textiles, sugar, tea/coffee processing and other agro-based industries.

Cultivation changes with:

- physical environment
- technological know-how
- socio-cultural practices

Primitive subsistence

small patches, simple tools, family/community labour, dependence on monsoon + natural fertility

Intensive subsistence

high population pressure, labour-intensive work, irrigation + biochemical inputs

Commercial farming

HYV seeds, chemical fertilisers, insecticides and pesticides for higher productivity

Plantation

single crop on large area, capital-intensive inputs, migrant labour, industry link

PRIMITIVE SUBSISTENCE FARMING

- Found in a few pockets of India; cultivated on small patches of land.
- Tools: hoe, dao and digging sticks; family/community labour.
- Depends on monsoon, natural soil fertility and environmental suitability.
- It is slash-and-burn farming: clear a patch, grow cereals/food crops, then shift when soil fertility declines.
- Shifting lets nature replenish fertility, but productivity is low because fertilisers and modern inputs are not used.

INDIAN NAMES

North-east: Jhumming; Manipur: Pamlou; Bastar district (Chhattisgarh): Dipa; also in Andaman and Nicobar Islands.

MORE INDIAN NAMES

Bewar/Dahiya (MP), Podu/Penda (AP), Pama Dabi/Koman/Bringa (Odisha), Kumari (Western Ghats), Valre/Waltre (SE Rajasthan), Khil (Himalayan belt), Kuruwa (Jharkhand).

SLASH & BURN: WORLD NAMES

Mexico + Central America

Milpa

Venezuela

Conuco

Brazil

Roca

Central Africa

Masole

Indonesia

Ladang

Vietnam

Ray



SHIFT → FRESH PATCH

RINJHA STORY

Rinjha lived near Diphu, Assam. Her family cleared/slashed/burned land and used a bamboo canal from a nearby spring. Declining soil fertility meant searching for a fresh patch next season.

INTENSIVE SUBSISTENCE FARMING

Practised in areas with high population pressure on land. It is labour-intensive and uses high doses of biochemical inputs plus irrigation for higher production.

- Inheritance divides holdings among generations and may make farm sizes uneconomical.
- Farmers still seek maximum output from limited land when alternative livelihoods are unavailable.
- This creates enormous pressure on agricultural land.

NCERT CHECK-POINT

The chapter asks you to name Indian states/areas where intensive subsistence farming is practised.

COMMERCIAL FARMING + PLANTATION

- Uses higher doses of modern inputs such as HYV seeds, chemical fertilisers, insecticides and pesticides.
- Commercialisation varies by region: rice is commercial in Haryana/Punjab but a subsistence crop in Odisha.

PLANTATION = COMMERCIAL FARMING

- single crop over a large area
- large tracts + capital-intensive inputs
- migrant labour
- produce used as industrial raw material
- tea, coffee, rubber, sugarcane and banana are important plantation crops
- tea in Assam/North Bengal; coffee in Karnataka
- good transport + communication link plantation, processing and markets



Banana plantation



Bamboo plantation

3. CROPPING PATTERN

India has diverse physical conditions and cultures, reflected in agricultural practices. Important crop groups include food/fibre crops, vegetables, fruits, spices and condiments. There are three main seasons: rabi, kharif and zaid.

RABI

Sown Oct–Dec; harvested Apr–Jun. Crops: wheat, barley, peas, gram, mustard. Important production in Punjab, Haryana, Himachal Pradesh, J&K, Uttarakhand and Uttar Pradesh. Winter precipitation from western temperate cyclones helps. Green Revolution boosted rabi crops in Punjab, Haryana, western UP and parts of Rajasthan.

KHARIF

Grown with the onset of monsoon; harvested Sep–Oct. Crops: paddy, maize, jowar, bajra, tur, moong, urad, cotton, jute, groundnut, soyabean. Major rice regions include Assam, West Bengal, coastal Odisha, Andhra Pradesh, Telangana, Tamil Nadu, Kerala, Maharashtra/Konkan, Uttar Pradesh and Bihar. Assam, West Bengal and Odisha can grow Aus, Aman and Boro rice.

ZAID

Short summer season between rabi and kharif. Watermelon, muskmelon, cucumber, vegetables and fodder are examples. Sugarcane takes almost a year.

4. MAJOR CROPS: RICE + WHEAT

RICE

- Staple food crop for a majority of Indians.
- Second largest rice producer in the world after China.
- Kharif; needs $>25^{\circ}\text{C}$, high humidity and $>100\text{ cm}$ annual rainfall.
- Lower-rainfall areas can grow it with irrigation.
- Plains of north/north-east India, coastal areas and deltaic regions are important.
- Canals + tubewells made rice possible in lower-rainfall Punjab, Haryana, western UP and parts of Rajasthan.



Rice cultivation

WHEAT

- Second most important cereal and a main food crop of north/north-west India.
- Rabi; cool growing season + bright sunshine at ripening.
- Needs $50\text{--}75\text{ cm}$ annual rainfall, evenly distributed during growth.
- Two main zones: Ganga-Satluj plains in the north-west and Deccan black-soil region.
- Major states: Punjab, Haryana, Uttar Pradesh, Bihar and Rajasthan.



Wheat crop



MILLET FAMILY + MAIZE

MILLETS

Jowar, bajra and ragi are important millets. Though called coarse grains, they are nutritionally valuable.

- Ragi is rich in iron, calcium, other micronutrients and roughage.
- Jowar is the third most important food crop by area and production; mostly rain-fed and needs little irrigation.
- Jowar states: Maharashtra, Karnataka, Andhra Pradesh, Madhya Pradesh.
- Bajra grows on sandy and shallow black soils; states: Rajasthan, Uttar Pradesh, Maharashtra, Gujarat, Haryana.
- Ragi is a dry-region crop; grows on red, black, sandy, loamy and shallow black soils; states include Karnataka, Tamil Nadu, HP, Uttarakhand, Sikkim, Jharkhand and Arunachal Pradesh.



MAIZE

- Used as food and fodder; mainly kharif.
- Needs 21–27°C and grows well in old alluvial soil.
- Bihar also grows maize in rabi season.
- HYV seeds, fertilisers and irrigation increased production.
- Major states: Karnataka, Madhya Pradesh, Uttar Pradesh, Bihar, Andhra Pradesh, Telangana.



Maize cultivation



Field crop

PULSES + SUGARCANE

PULSES

- India is the largest producer and consumer of pulses in the world.
- Major vegetarian source of protein.
- Main pulses: tur/arhar, urad, moong, masur, peas, gram.
- Need less moisture and survive dry conditions.
- All except arhar help restore soil fertility by nitrogen fixation.
- Often grown in rotation with other crops.
- Major states: Madhya Pradesh, Rajasthan, Maharashtra, Uttar Pradesh, Karnataka.

SEASON CHECK

The chapter asks you to distinguish which listed pulses belong to kharif and which to rabi.



Pulse crop

SUGARCANE

- Tropical + sub-tropical crop; hot and humid climate.
- Needs 21–27 °C and 75–100 cm annual rainfall; irrigation where rainfall is low.
- Can grow on a variety of soils; manual labour needed from sowing to harvesting.
- India is second largest producer after Brazil.
- Main source for sugar, gur/jaggery, khandsari and molasses.
- Major states: Uttar Pradesh, Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, Telangana, Bihar, Punjab, Haryana.



OIL SEEDS + TEA

OIL SEEDS

- In 2020, India was the second largest groundnut producer after China.
- Oilseeds occupy about 12% of total cropped area.
- Groundnut, mustard, coconut, sesamum/til, soyabean, castor, cotton seed, linseed, sunflower are major oilseeds.
- Most are edible; some are raw material for soap, cosmetics and ointments.
- Groundnut is kharif and about half of major oilseeds; Gujarat led in 2019-20, then Rajasthan and Tamil Nadu.
- Linseed + mustard = rabi; sesamum = kharif in north/rabi in south; castor = both.



Field produce

TEA

- Plantation agriculture + important beverage crop; introduced initially by the British.
- Most tea plantations today are owned by Indians.
- Needs tropical/sub-tropical climate, deep fertile well-drained humus-rich soil, warm/moist frost-free conditions.
- Frequent evenly distributed showers support continuous tender-leaf growth.
- Labour-intensive: abundant, cheap and skilled labour needed.
- Processed within the tea garden to retain freshness.
- Major states: Assam, Darjeeling/Jalpaiguri hills (WB), Tamil Nadu, Kerala; also HP, Uttarakhand, Meghalaya, AP, Tripura.
- In 2020, India ranked second in tea production after China.



Tea plantation



Tea leaf harvesting

COFFEE + HORTICULTURE CROPS

COFFEE

- Indian coffee is known worldwide for good quality.
- Arabica was first brought from Yemen; it has high world demand.
- Cultivation began on Baba Budan Hills and is now concentrated in Karnataka, Kerala and Tamil Nadu; the chapter notes the Nilgiri region.



Coffee plantation



Fruit crop



Vegetable crop

HORTICULTURE

In 2020, India was the second largest producer of fruits and vegetables after China. It produces tropical as well as temperate fruits.

- Mango – Maharashtra, Andhra Pradesh, Telangana, Uttar Pradesh, West Bengal.
- Orange – Nagpur + Cherrapunjee (Meghalaya).
- Banana – Kerala, Mizoram, Maharashtra, Tamil Nadu.
- Litchi + guava – Uttar Pradesh and Bihar.
- Pineapple – Meghalaya; grapes – Andhra Pradesh, Telangana, Maharashtra.
- Apples, pears, apricots, walnuts – J&K and Himachal Pradesh.
- India also produces peas, cauliflower, onion, cabbage, tomato, brinjal and potato.

MARKET NOTE

Horticultural products are in demand widely, so access to transport, storage and markets matters.

5. NON-FOOD CROPS: COTTON + RUBBER + JUTE

COTTON

- India is believed to be the original home of cotton; it is a major raw material for cotton textiles.
- India is the second largest cotton producer after China.
- Best in drier parts of Deccan black cotton soil.
- Needs high temperature, light rainfall/irrigation, about 210 frost-free days and bright sunshine.
- Kharif crop; matures in 6-8 months.
- Major states in the chapter: Maharashtra, Gujarat, Madhya Pradesh, Karnataka, Andhra Pradesh, Telangana, Tamil Nadu, Punjab, Haryana, Uttar Pradesh.



Cotton cultivation

RUBBER

- Equatorial crop; also grown in tropical/sub-tropical areas under special conditions.
- Needs moist humid climate, >200 cm rain and temperature above 25°C.
- Industrial raw material; important areas: Kerala, Tamil Nadu, Karnataka, Andaman & Nicobar, Garo hills (Meghalaya).

JUTE

- Known as the golden fibre; grows on fertile well-drained floodplain soils renewed yearly.
- Needs high temperature during growth.
- Major states: West Bengal, Bihar, Assam, Odisha, Meghalaya.
- Used for gunny bags, mats, ropes, yarn, carpets and other artefacts.

FIBRE CROPS + TECHNOLOGICAL / INSTITUTIONAL REFORMS

FIBRE CROPS

Cotton, jute, hemp and natural silk are the four major fibre crops. The first three are derived from crops grown in soil; silk comes from silkworm cocoons fed on mulberry leaves. Rearing silkworms for silk fibre is called sericulture.

WHY REFORMS?

Agriculture has been practised in India for thousands of years. Sustained land use without compatible techno-institutional change slowed development. Even with irrigation, many farmers still depended on monsoon and natural soil fertility. Population growth made this a serious challenge.

AFTER INDEPENDENCE

- Agriculture provides livelihood to more than 60% of the population, so technical and institutional reform is important.
- Collectivisation, consolidation of holdings, cooperation and abolition of zamindari were important institutional reforms.
- Land reform was the main focus of the First Five Year Plan.
- Inheritance caused fragmentation of landholdings; consolidation was needed. Laws were enacted but implementation was weak/lukewarm.
- 1960s–70s: Green Revolution (package technology) + White Revolution (Operation Flood).
- Because benefits concentrated in selected areas, land development programmes in the 1980s–90s combined institutional + technical reform.

REFORMS + SUPPORT MEASURES FOR FARMERS

LAND + INSTITUTIONAL

Collectivisation | consolidation of holdings | cooperation | abolition of zamindari | land reform | tackling fragmentation

TECHNICAL + FINANCIAL

Crop insurance against drought, flood, cyclone, fire and disease. Grameen banks, cooperative societies and banks provide loans at lower interest.

NAMED SCHEMES

Kisan Credit Card (KCC) and Personal Accident Insurance Scheme (PAIS) were introduced for farmers.

GREEN + WHITE REVOLUTION

Green Revolution = package technology/modern inputs for crop production. White Revolution = Operation Flood.



Modern agricultural equipment

MARKET + INFORMATION SUPPORT

- Special weather bulletins + agricultural programmes on radio and television.
- Government announces minimum support, remunerative and procurement prices for important crops.
- These measures help protect farmers from exploitation by speculators and middlemen.
- The chapter also directs students to the Farmers' Portal for information on agriculture, horticulture and schemes.

6. BHOODAN – GRAMDAN: "BLOOD-LESS REVOLUTION"

- Vinoba Bhave was regarded by Gandhi as his spiritual heir; he supported gram swarajya and was a foremost satyagrahi.
- After Gandhi's martyrdom, Vinoba undertook a padyatra across the country to spread Gandhi's message.
- At Pochampally, Telangana, poor landless villagers asked for land for economic well-being.
- Vinoba said he would approach the Government if they undertook cooperative farming.
- Shri Ram Chandra Reddy offered 80 acres for distribution among 80 landless villagers.

BHOODAN

Donation of land for distribution among landless villagers became known as Bhoodan.

GRAMDAN

Later some zamindars/large owners offered villages for the landless; this was called Gramdan. Some landowners also donated land because of fear of the land-ceiling law.

KEY TAKEAWAY

The Bhoodan-Gramdan movement initiated by Vinoba Bhave is described as the Blood-less Revolution.



Pochampally 80 acres Bhoodan Gramdan Spread
landless demanded Reddy offers land land donation village donation movement across India

7. NCERT EXERCISES – QUICK REVISION SHEET

MCQs

1. Single crop on a large area?
Key: Plantation agriculture
2. Which is a rabi crop?
Key: Gram
3. Which is leguminous?
Key: Pulses

30-WORD ANSWERS TO PREPARE

- One important beverage crop + geographical conditions.
- One staple crop of India + regions where it is produced.
- Institutional reform programmes introduced for farmers.

120-WORD ANSWERS

- Government initiatives taken to increase agricultural production.
- Geographical conditions required for rice.

PROJECT WORK

- Group discussion: necessity of literacy among farmers.
- On an outline map of India, show wheat-producing areas.

PORTAL TASK

Collect information about agriculture, horticulture and agricultural schemes from the Farmers' Portal and discuss the benefits of the information.

EXAM TIP

Remember: crop + season + temperature/rainfall + soil + major states. This pattern works for most crop questions.

MAP PRACTICE

Rice and wheat distribution maps are part of the chapter. Practise locating important producing regions.

8. ACTIVITY + ONE-PAGE RECAP

WORD-PUZZLE CLUES

1. Two staple food crops of India
Key from clue: Rice + Wheat
2. Summer cropping season
Key from clue: Zaid
3. Pulses contain...
Key from clue: Protein
4. A coarse grain
Key from clue: Ragi
5. Two important beverages
Key from clue: Tea + Coffee
6. One major fibre on black soils
Key from clue: Cotton

ULTIMATE RECAP

Farming: Primitive → Intensive → Commercial → Plantation

Seasons: Rabi (winter) | Kharif (monsoon) | Zaid (summer)

Food crops: Rice, wheat, millets, maize, pulses, sugarcane

Plantation: Tea, coffee, rubber, sugarcane, banana

Horticulture: Fruits + vegetables; tropical + temperate

Fibres: Cotton, jute, hemp, natural silk; sericulture

Reforms: Land reform + Green Revolution + White Revolution + insurance + credit + price support

Bhoodan: Vinoba Bhave → land donation → Gramdan → Blood-less Revolution